

Four Corners:

Analytical

Let $f(x) = 4 - 3 \cos(\pi x)$.

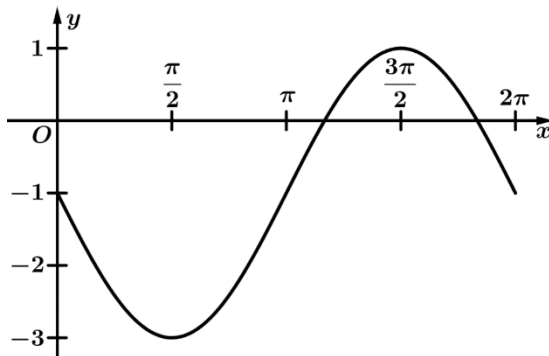
What is the maximum value of $f(x)$?

Numerical

The function g is defined by $g(x) = a \cos(2x) + d$, where a and d are constants. Selected values of g are shown in the table below. Find the missing values in the table.

x	0	$\pi/2$	$3\pi/4$	π	$3\pi/2$
$g(x)$	4		1		

Graphical



A portion of the graph of $h(x)$ is shown above, where $h(x) = a \sin(x) + d$. Find the values of the constants a and d .

Verbal

Let $k(x)$ be a sine function with period π , amplitude of 3, midline $y = -1$, and no other transformations of $y = \sin(x)$.

Write an equation for $k(x)$.

Big 10:

Complete the ten problems. Once you complete each problem, you will cross off one of the numbers in the box below. If done correctly, you will have all 10 numbers crossed off with no repeats.

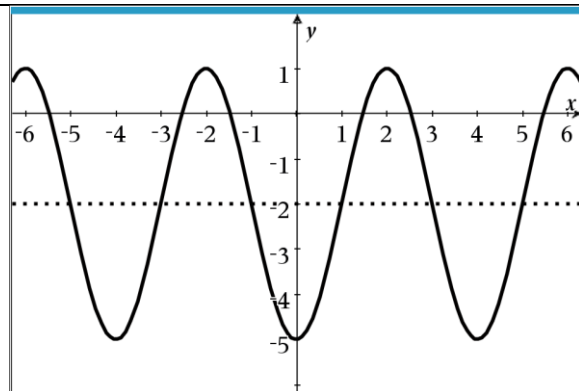
No Calculator

0	1	2	3	4	5	6	7	8	9
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Complete each problem.		Answer
A. The function $f(\theta) = \sin \theta$ is transformed to produce the graph of $g(\theta)$. What is the amplitude if $g(\theta) = -\sin(3\theta + \pi)$?	Amplitude is	
B. Find the period of the graph of the sinusoidal function $g(\theta) = 2\cos\left(\frac{\pi}{4}(\theta)\right) - 3$.	Period is	
C. The function $h(\theta) = \cos \theta$ is transformed to produce the graph of $j(\theta)$. What is the equation of the midline of the graph of the sinusoidal function $j(\theta) = 2\cos(4\theta - 3\pi) + 7$?	Midline is $y =$	
D. How many complete cycles does the sinusoid function $f(\theta) = 4\sin(9\theta) + 3$ have in the interval $[0, 2\pi]$?	Cycles =	
E. Find the frequency of the graph of the sinusoidal function $g(\theta) = 2\cos(10\pi(\theta)) - 3$.	Frequency is	

Use for F. and G.

In the figure to the right, the graph of a sinusoidal function of the form $g(\theta) = a \sin(b(\theta + c)) + d$, where a , b , c , and d are constants and $a > 0$. The graph of the midline, $y = -2$, is also shown as a dotted line.



F. What is the value of a ?	$a =$																			
G. What is the period of the sinusoidal function?	Period is																			
H. The function $h(\theta) = \cos \theta$ is transformed to produce the graph of $j(\theta)$. What is the horizontal translation (or phase shift) of the graph $h(\theta)$ in the sinusoidal function $j(\theta) = 3h(2\pi\theta + 4\pi) + 7$?	Circle one: Right Left																			
I. Selected values of a sinusoidal function of the form $g(\theta) = a \sin(b(\theta + c)) + d$ are given in the table to the right. It is known that a , b , c , and d are constants and $a \neq 0$. What is the value of d ?	<table><tr><td>θ</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$f(\theta)$</td><td>9</td><td>6</td><td>3</td><td>6</td><td>9</td><td>6</td><td>3</td><td>6</td></tr></table>	θ	-1	0	1	2	3	4	5	6	$f(\theta)$	9	6	3	6	9	6	3	6	$d =$
θ	-1	0	1	2	3	4	5	6												
$f(\theta)$	9	6	3	6	9	6	3	6												
J. The graph of a sinusoidal function has the form of $k(\theta) = a \cos(b(\theta + c)) + d$. If $k(\theta) = \frac{5}{\pi} \cos\left(\frac{\pi}{5}\theta - 1\right) + 8$, find a and c .	$a + c =$																			

Sinusoidal Function Equations Practice:

A. Answer the questions about each sinusoidal function.

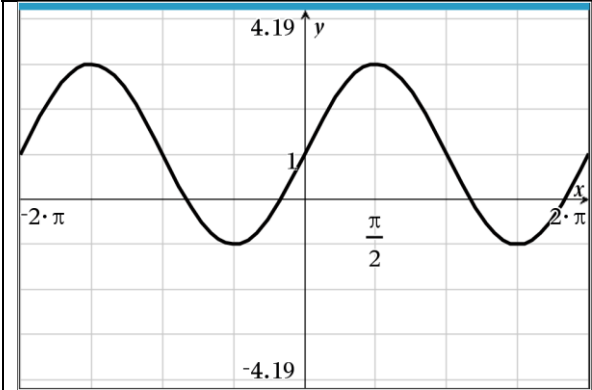
1. Transformations in the following order are performed on the graph of $f(\theta) = \cos \theta$ to produce the graph of $g(\theta)$. Write a function expression for $g(\theta)$.

A horizontal translation (phase shift) of $-\frac{\pi}{8}$ units, a horizontal dilation by a factor of 2 (period of 4π), a vertical dilation by a factor of 5 (amplitude of 5), and a vertical translation of 6 units.

θ	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{2}$
$f(\theta)$	1	-2	$4 - 3\sqrt{3}$	10

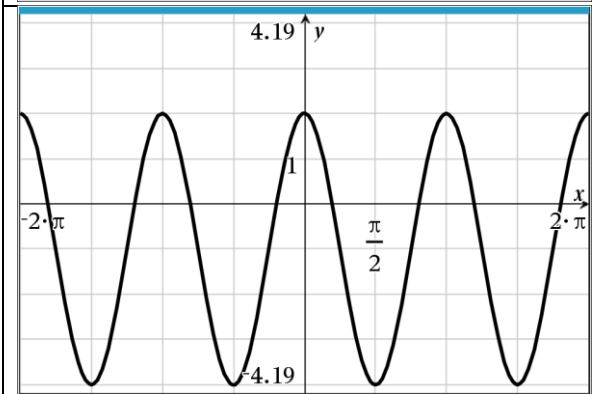
2. The function f can be modeled by a sine function of the form $f(\theta) = a \sin(\theta) + d$. Selected values of $f(\theta)$ are shown in the table above. What are the values of a and d for the function f ?

B. Write sine and cosine functions for the graphs shown.



1. The graph of a transformed **sine** function is shown to the right. The midline is $y = 1$. Write a possible function f for the graph shown.

2. Write the function in terms of cosine.



1. The graph of a transformed **cosine** function is shown to the right. The midline is $y = -1$. Write a possible function f for the graph shown.

2. Write the function in terms of sine.